



Week 01 – Task 02

Packet Analysis & Protocol Deep-Dive

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Date: 03/07/2026



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1. Objective

To use industry-standard tools to visualize how data moves across the wire. This task involves performing live packet analysis via Wireshark to observe network traffic, analyzing and documenting the TCP Three-Way Handshake (SYN, SYN-ACK, ACK) in a live capture, and identifying common protocol headers (Ethernet, IP, TCP) and their forensic significance.

2. Tools Used

Tool	Purpose
Wireshark	Live packet capture and protocol analysis
Netcat (nc)	Generate TCP connections for handshake capture
Kali Linux	Attacker machine – all tools run here
Loopback (lo)	Interface used for local TCP traffic capture

3. TCP Three-Way Handshake – Theory

The TCP Three-Way Handshake is the process used to establish a reliable connection between two hosts before any data is exchanged. It consists of three steps:

Step	Flag	Description
Step 1	SYN	Client sends synchronization request to server
Step 2	SYN-ACK	Server acknowledges and sends its own SYN
Step 3	ACK	Client confirms – connection established

4. Lab Setup – Generating TCP Traffic

4.1 Netcat Server (Terminal 1)

A Netcat listener was started on port 4444 to act as the TCP server:

```
$ nc -lvp 4444
listening on [any] 4444 ...
connect to [127.0.0.1] from localhost [127.0.0.1] 34268
```

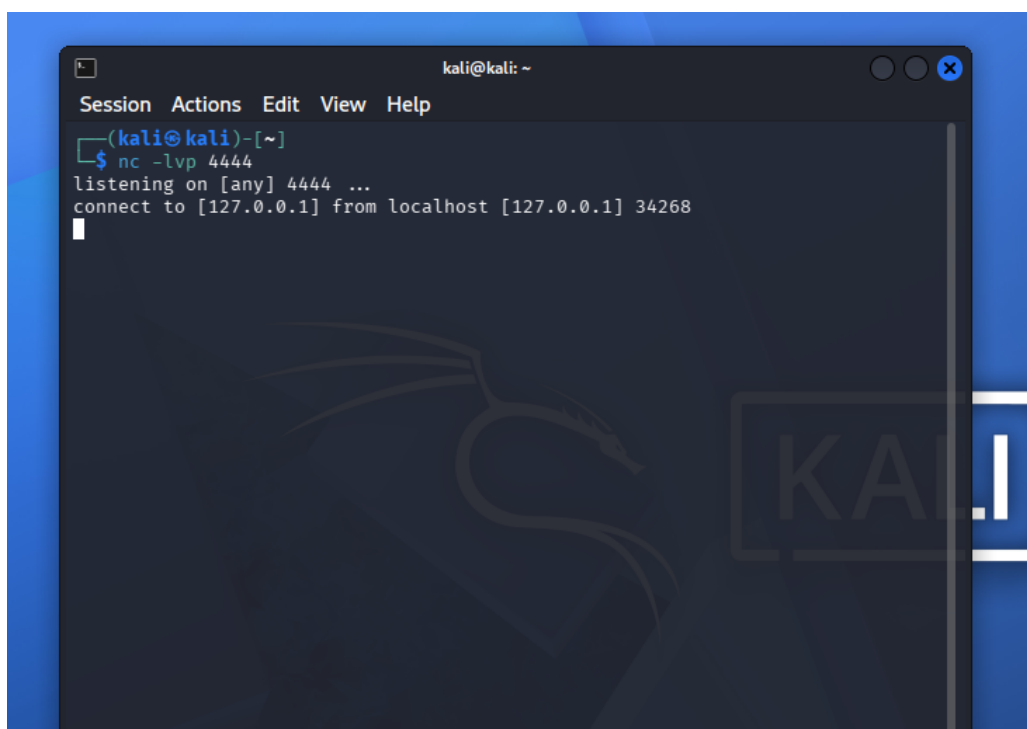


Figure 1: Netcat server listening on port 4444 – connection received

4.2 Netcat Client (Terminal 2)

A second terminal was used to initiate a TCP connection to the listener:

```
$ nc 127.0.0.1 4444
```

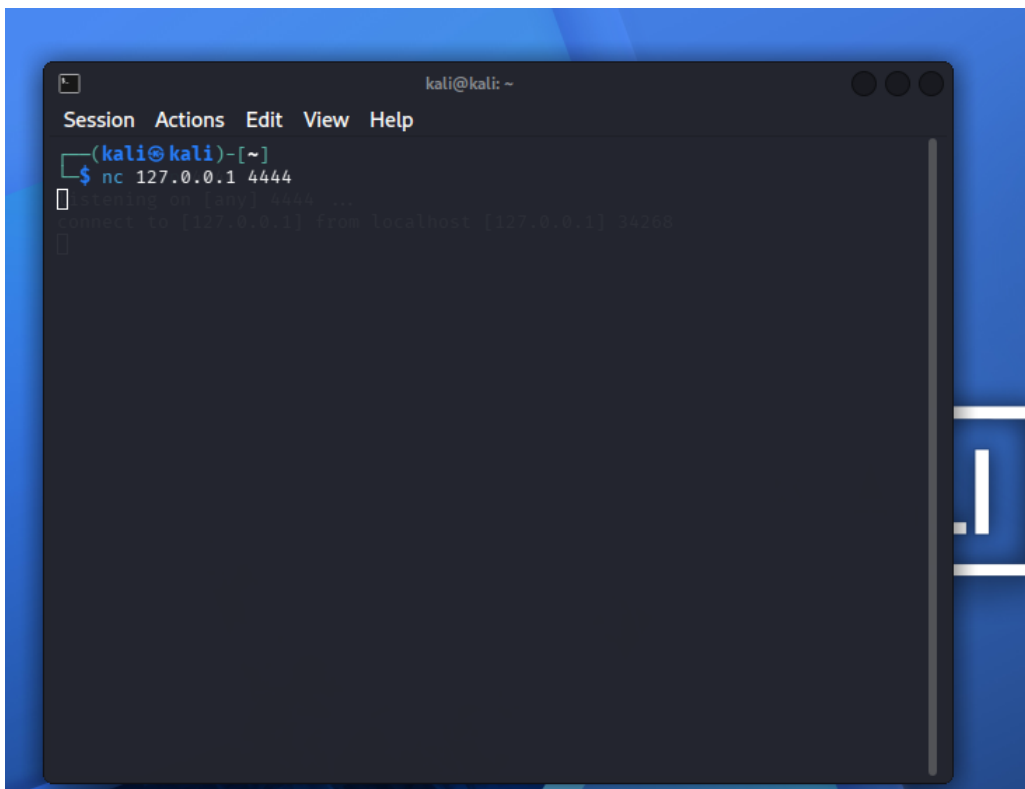


Figure 2: Netcat client connecting to 127.0.0.1 port 4444

5. Wireshark – Live Packet Capture

5.1 Capture Overview

Wireshark was configured to capture on the **Loopback (lo)** interface with the filter `tcp.port==4444`. A total of **6 packets** were captured showing the complete TCP session including the Three-Way Handshake.

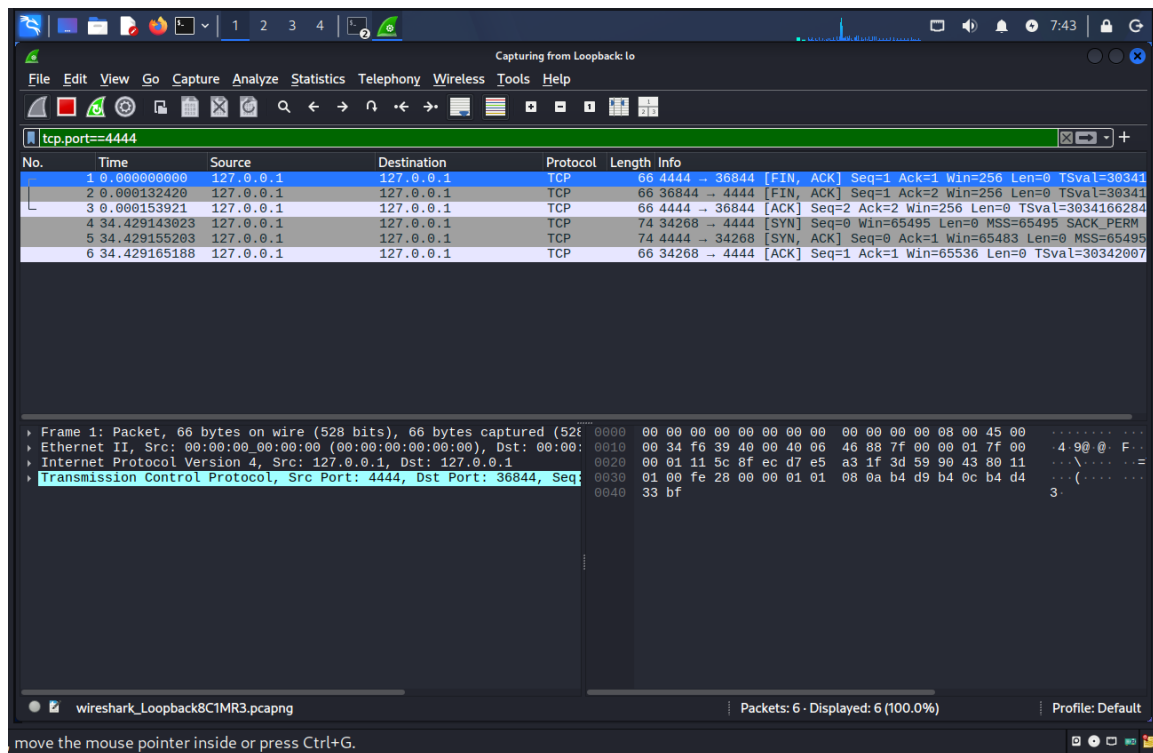


Figure 3: Wireshark capture – 6 TCP packets on loopback interface

Pkt	Source	Destination	Flag	Description
1	127.0.0.1	127.0.0.1	FIN, ACK	Previous session close
2	127.0.0.1	127.0.0.1	FIN, ACK	Previous session close
3	127.0.0.1	127.0.0.1	ACK	Previous session ACK
4	127.0.0.1	127.0.0.1	SYN	Handshake Step 1
5	127.0.0.1	127.0.0.1	SYN-ACK	Handshake Step 2
6	127.0.0.1	127.0.0.1	ACK	Handshake Step 3

6. TCP Three-Way Handshake Analysis

6.1 Step 1 – SYN Packet (Packet 4)

The client (port 34268) sends a SYN packet to the server (port 4444) to initiate the connection. Sequence Number is 0 (relative).

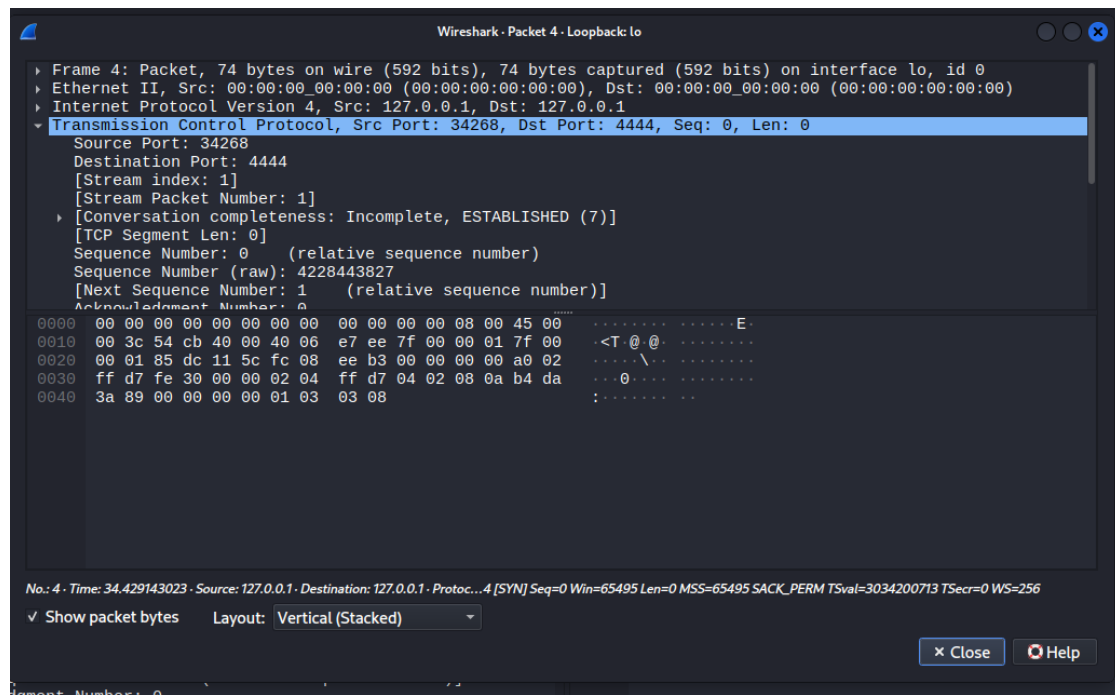


Figure 4: Packet 4 – SYN flag set, Seq=0, initiating connection.

Source Port:	34268
Destination Port:	4444
Flag:	SYN (0x002)
Sequence Number:	0 (relative)
Ack Number:	0
Window Size:	65495

6.2 Step 2 – SYN-ACK Packet (Packet 5)

The server (port 4444) responds with SYN-ACK, acknowledging the client's SYN and sending its own synchronization request. Seq=0, Ack=1.

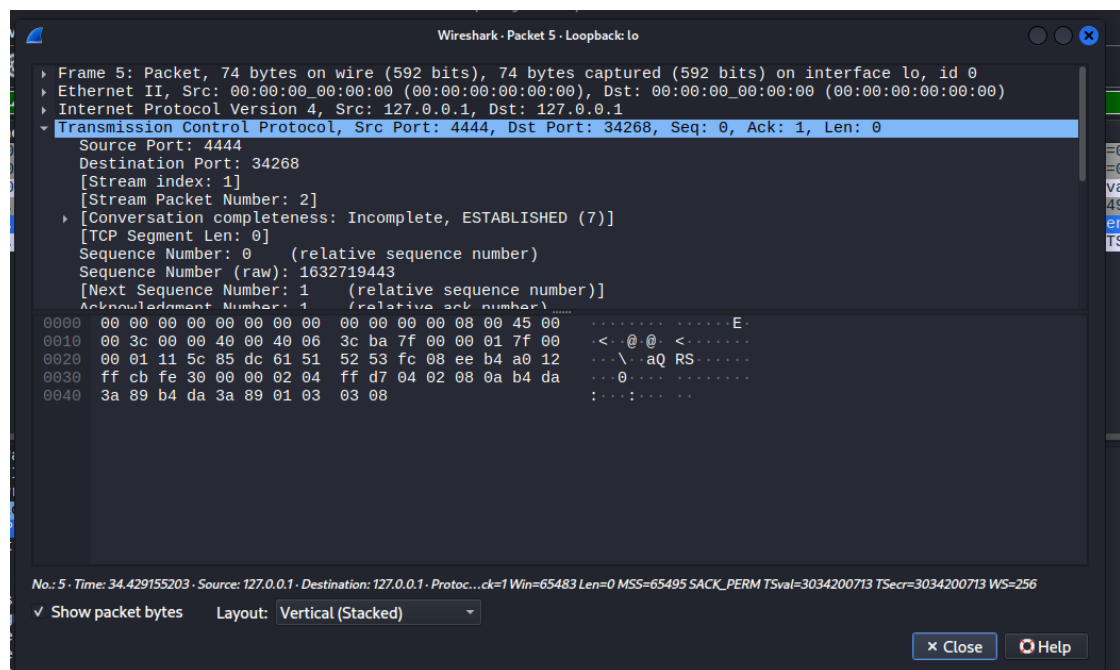
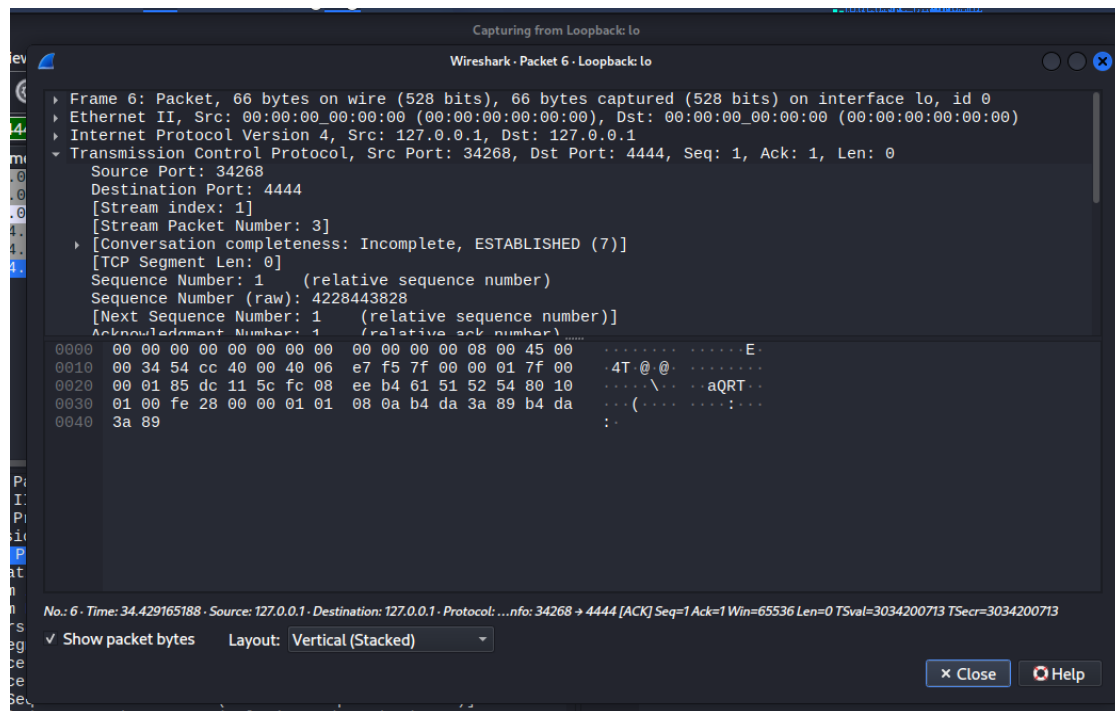


Figure 5: Packet 5 – SYN-ACK flag set, Seq=0, Ack=1

Source Port:	4444
Destination Port:	34268
Flag:	SYN-ACK (0x012)
Sequence Number:	0 (relative)
Ack Number:	1
Window Size:	65483

6.3 Step 3 – ACK Packet (Packet 6)

The client sends final ACK to complete the handshake. Connection is now **ESTABLISHED**. Seq=1, Ack=1.



Source Port:	34268
Destination Port:	4444
Flag:	ACK (0x010)
Sequence Number:	1
Ack Number:	1
Window Size:	65536

7. Protocol Header Analysis

7.1 All Three Headers Expanded

The screenshot below shows all three protocol layers expanded for the ACK packet, demonstrating the encapsulation of Ethernet, IP, and TCP headers.

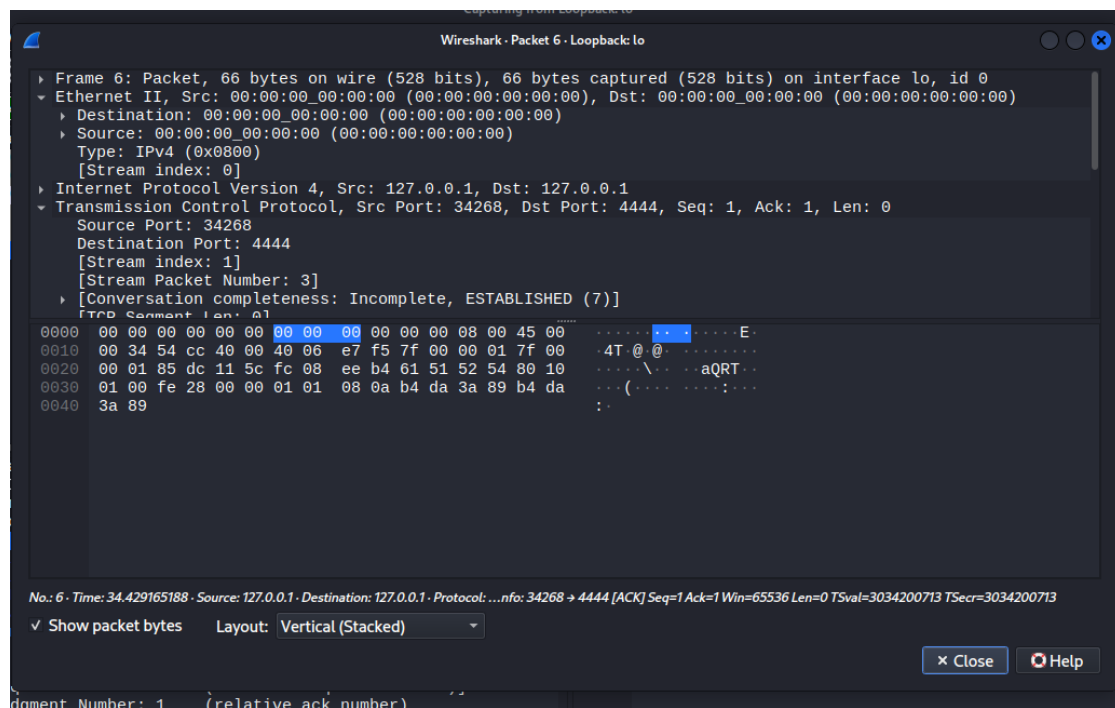


Figure 7: Ethernet II, IP, and TCP headers expanded – ACK packet

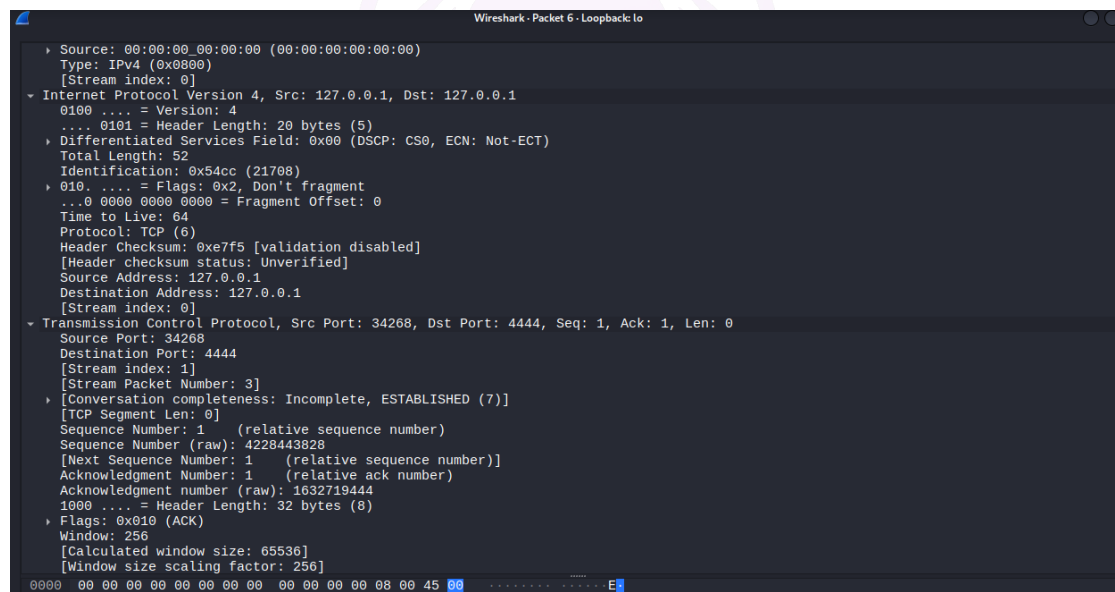


Figure 8: IP Header details – Version 4, TTL=64, Protocol=TCP

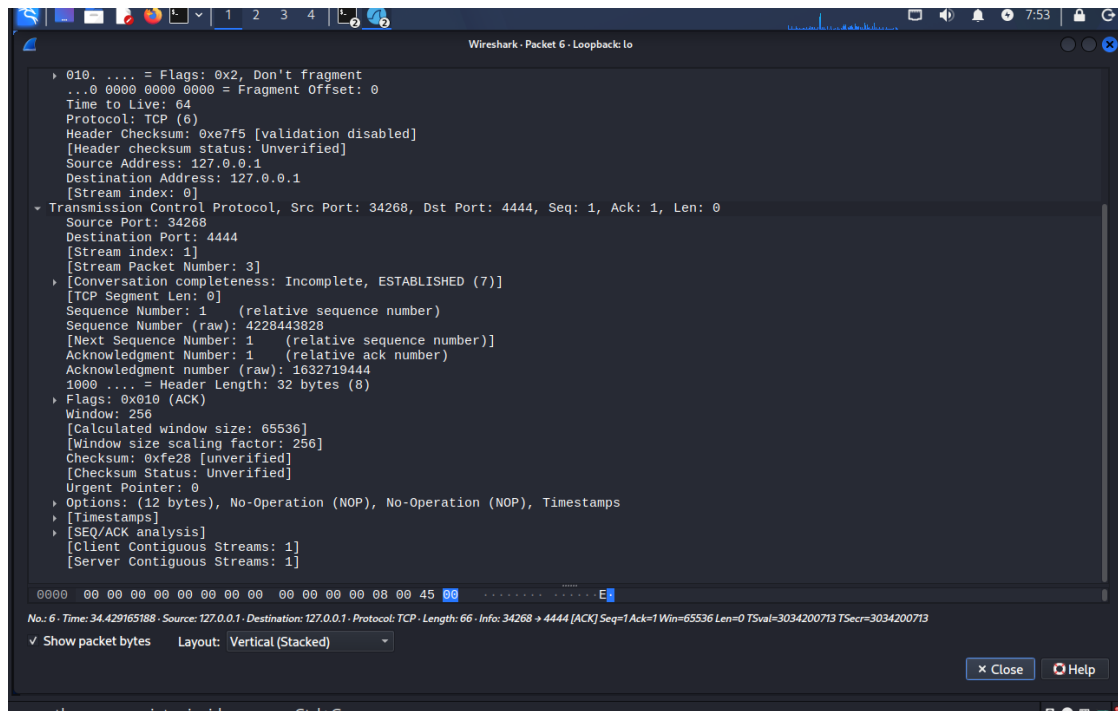


Figure 9: TCP Header details – Flags, Sequence Numbers, Window Size

7.2 Protocol Header Summary

Header	Key Fields	Forensic Significance
Ethernet II	MAC Address, Type	Identifies physical device, Layer 2 communication
Internet Protocol v4	Src/Dst IP, TTL, Protocol	Tracks source & destination, hop count
TCP	Ports, Flags, Seq/Ack	Shows connection state, detects port scans

8. Results Summary



Task	Status	Details
Wireshark Installed & Launched	DONE	Kali Linux
Loopback Interface Captured	DONE	lo interface
TCP Traffic Generated	DONE	Netcat port 4444
SYN Packet Captured	PASS	Packet 4
SYN-ACK Packet Captured	PASS	Packet 5
ACK Packet Captured	PASS	Packet 6
Ethernet Header Analyzed	DONE	MAC, Type
IP Header Analyzed	DONE	TTL, Protocol
TCP Header Analyzed	DONE	Flags, Seq/Ack

9. Learning Outcomes



1. Mastered the use of **Wireshark** to perform live packet capture on the loop-back interface and apply display filters effectively.
2. Successfully captured and analyzed the complete **TCP Three-Way Handshake** (SYN, SYN-ACK, ACK) using Netcat to generate controlled traffic.
3. Understood the role of each TCP flag and how Sequence and Acknowledgment numbers establish a reliable connection.
4. Identified and analyzed **Ethernet, IP, and TCP protocol headers**, understanding their structure and forensic significance in network analysis.
5. Gained foundational skills in protocol analysis essential for detecting network anomalies, port scans, and malicious traffic in future Purple Team exercises.

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